

YERI PARK

Embedded ML Systems Engineer

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Research Interests:

- Computer Architecture (Software–Hardware Co-Design, Compiler Design, Heterogeneous Memory Management)
- Embedded ML Systems (RTOS, Real-Time Inference, Deployable Edge AI)
- Efficient AI (Quantization, Pruning, Knowledge Distillation, KV Cache Optimization)

EXPERIENCE

Hyundai Motor Company

Seoul, Republic of Korea

Researcher, Department of Autonomous Driving SW Development

Jan 2022 – Present

- Deployed mixed-precision quantization (PTQ/QAT), knowledge distillation, and structured pruning for production autonomous-driving DNNs, reducing inference latency and memory footprint on NVIDIA Jetson(Thor/Orin) and Qualcomm Snapdragons; built the validation framework confirming recovered full-precision accuracy
- Optimized VLM/VLA inference across GPU server clusters and edge devices via automated layer-wise quantization and host ↔ GPU/NPU memory allocation, cutting end-to-end inference overhead
- Developed a QNX/RTOS-based embedded AI inference framework using C/C++, CUDA, and OpenCL for autonomous systems, integrating car-network interfaces (LIN, CAN, Ethernet, LVDS), multimodal camera/LiDAR/Radar sensor logging, Rust-based visualization, and HILS/SILS validation across NVIDIA and Qualcomm platforms
- Built a DNN optimization automation framework for GPU-cluster host ↔ device memory optimization, inference acceleration, and validation testing against original DNN artifacts

Korea Institute of Science and Technology

Seoul, Republic of Korea

Undergraduate Research Intern, Robot Vision Lab

Sep 2021 – Dec 2021

- Designed RGB-D CNN road segmentation and freespace perception for the robot-vision of wearable gait-assist robots on NVIDIA Xavier, and developed ROS-based software for RGB-D camera and LiDAR data logging alongside real-time visualization

Hyundai Motor Company

Seoul, Republic of Korea

Undergraduate Research Intern, LiDAR Perception Development

Jun 2019 – Aug 2019

- Built LiDAR sparse point cloud segmentation DNN, preprocessing, ground segmentation, and point-cloud clustering pipelines; implemented a real-time PCD visualization tool

Ewha Womans University

Seoul, Republic of Korea

Undergraduate Research Intern, Digital Signal Processing Lab

Sep 2021 – Dec 2021

- Developed a human respiration detection CNN model using IR-UWB radar signals

EDUCATION

Georgia Institute of Technology

Atlanta, United States

Master of Science in Computer Science

Sep 2025 – Present

Ewha Womans University

Seoul, Republic of Korea

Bachelor of Science in Electronic and Electrical Engineering, GPA: 97.9/100

Mar 2017 – Feb 2022

SKILLS

Programming: C/C++, Python (PyTorch), Rust, CUDA/OpenCL, Haskell

Embedded & RTOS: FreeRTOS, Linux Kernel, Yocto Project, HILS/SILS Validation

Automotive Protocols: CAN, LIN, Automotive Ethernet, LVDS

ML Systems: TensorRT, ONNX-MLIR, PTQ/QAT, Knowledge Distillation, Pruning, Mixed-Precision Quantization, KV Cache Optimization

Platforms: NVIDIA Jetson(Thor, Orin, Xavier), Qualcomm Snapdragon, NVIDIA A100/H100/H200

Certifications: ISTQB CTFL, ISTQB AI Testing, NVIDIA CUDA Programming

Languages: Korean (Native), English (Fluent)

ADDITIONAL PROJECTS

Edge AI Network Pruning (Aug 2025) — Applied structured/unstructured pruning to CV networks on NVIDIA Orin. C/C++, TensorRT-LLM, Swin Transformer

Rust-Based ONNX-MLIR Optimization (Apr 2025) — Built a network optimization framework for universal hardware compilation. Rust, Python, C++, ONNX-MLIR

Autonomous Driving LiDAR Perception (Dec 2023) — Ground segmentation, curb detection, free-space estimation, 3D object detection. C/C++, CUDA, OpenCL, PyTorch

HONORS & AWARDS

2nd Prize, Ewha Engineering Capstone Design Contest (2020) — AI-based audience response analysis system

Excellence Award, Korea Electric Power Engineering Camp (2020) — Top 3, national engineering camp

Academic Excellence Scholarship, Ewha Womans University — 4 consecutive semesters (2017–2021)